ALHASSAN AHMED

AI Engineer | Data Scientist

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PROFESSIONAL SUMMARY

AI Engineer and Data Scientist (B.Sc. in Computer Science, 2026) specializing in building and shipping end-to-end AI systems. Proficient in predictive modeling, NLP, computer vision, and LLM/RAG applications, with strong expertise in on-device deployment using ONNX Runtime, FastAPI, and Docker. Skilled at converting raw sensor, financial, and text data into deployed, measurable products.

TECHNICAL SKILLS

Programming Languages: Python (OOP, Data Structures), SQL, Java, C++, C#, JavaScript, TypeScript, Kotlin

Machine Learning & Deep Learning: Scikit-learn, PyTorch, TensorFlow, Keras, CNNs, ANNs, Computer Vision (OpenCV), Predictive Modeling, Classification, Regression, Clustering, Time Series Analysis & Forecasting, Knowledge Distillation, Model Evaluation (Precision, Recall, F1-Score, ROC-AUC), Imbalanced Data Handling

NLP & LLMs: Natural Language Processing, TF-IDF, Sentiment Analysis, Large Language Models (Llama 3.3 70B), Retrieval-Augmented Generation (RAG), FAISS Vector Search, Prompt Engineering, LLM Safety (Llama Guard)

Data Analysis & Visualization: Pandas, NumPy, Data Preprocessing, Feature Engineering, Exploratory Data Analysis (EDA), Matplotlib, Seaborn, Power BI, Interactive Dashboards

Deployment & MLOps: FastAPI, Flask, RESTful APIs, Docker, ONNX Runtime (on-device inference), Hugging Face Spaces, Git, GitHub

Databases & Backend: SQL Server, MongoDB, PostgreSQL (Supabase), ASP.NET Core MVC, Entity Framework Core

PROJECTS

Smart Health Monitoring System — *Graduation Project, Future University in Egypt* *Spring 2026*

End-to-end AI health platform featuring on-device deep learning and an LLM/RAG clinical assistant.

- **On-Device Deep Learning:** Developed a dual-stream 1D-CNN (“FusionNet”) for fall detection achieving 0.971 macro-AUC, and a dual-head model for activity recognition reaching 94.4% accuracy on the WISDM benchmark.
- **Model Compression & Edge AI:** Compressed a cardiac beat classifier to 64 KB via knowledge distillation (0.96 N-recall) and deployed models with ONNX Runtime for privacy-preserving, sub-10 ms offline inference.
- **LLM/RAG Architecture:** Engineered a bilingual RAG clinical assistant (Llama 3.3 70B, FAISS) with drug-interaction detection, achieving a 100% pass rate on a strict 64-prompt medical safety evaluation.
- **Hardware Integration:** Validated the end-to-end pipeline on Wear OS (Samsung Galaxy Watch 5), successfully triggering physical fall SOS alerts backed by a defense-in-depth safety architecture.
- **Tech Stack:** *Python, PyTorch, ONNX Runtime, FAISS, Groq, FastAPI, Supabase, React Native, Wear OS, Docker.*

Credit Risk & Loan Approval Prediction | *Python, Pandas, Scikit-learn*

- Predicted loan approval outcomes and quantified credit risk from customer financial data by engineering features, encoding categorical variables, and benchmarking Scikit-learn classifiers with ROC-AUC and F1-score.

Fraud Detection System | *Python, Scikit-learn*

- Flagged fraudulent transactions in a highly imbalanced dataset by applying resampling strategies and optimizing the precision–recall trade-off to minimize missed fraud cases.

Customer Churn Prediction | *Python, Pandas, Scikit-learn*

- Identified customers at risk of churning from behavioral and transactional data by training and comparing classification models, enabling targeted retention decisions.

NLP-Based Customer Feedback Analysis | *Python, NLP, TF-IDF*

- Extracted sentiment and recurring complaint themes from customer reviews by building an NLP pipeline with text preprocessing and TF-IDF vectorization, turning unstructured feedback into ranked, actionable insights.

Diabetes Risk Prediction | *Python, Scikit-learn*

- Predicted diabetes risk from patient health records through a complete ML workflow: data cleaning, feature engineering, model training, and evaluation with precision, recall, and ROC-AUC.

ML Model Deployment (REST API) | *FastAPI, Flask, Docker*

- Served real-time predictions from a trained machine learning model by containerizing a FastAPI/Flask REST API with Docker, demonstrating a production-style deployment workflow.

Business Performance Dashboard | *Power BI, SQL, Python*

- Tracked revenue, profit, and core business KPIs in interactive Power BI dashboards by modeling data with SQL and Python, giving stakeholders a single self-serve view of performance.

EXPERIENCE & ACHIEVEMENTS

Data Science & AI Trainee — *Instant* *2026*

- Prepared modeling-ready datasets by performing data cleaning, preprocessing, and feature engineering with Pandas and NumPy.
- Trained and fine-tuned machine learning and deep learning models, evaluating them with precision, recall, F1-score, and ROC-AUC to select the best performers.
- Built NLP models for sentiment extraction and delivered interactive dashboards that communicated predictions for business and healthcare use cases.

RoboCup Junior — Rescue Robotics — *International Competition* *Germany | 2024*

- **Won 2nd place in Egypt and 1st and 3rd place finishes in the European rounds** by building autonomous rescue robots in Python with OpenCV and NumPy.
- Improved real-time victim-detection performance by optimizing computer-vision pipelines under strict on-board compute constraints.
- Mentored new team members and presented solutions at international workshops, strengthening team output across competition seasons.

Backend Developer — *Academic Projects*

- Shipped full-stack web applications with ASP.NET Core MVC and C#, exposing ML/AI model outputs through RESTful APIs.
- Reduced query response times by optimizing database access across MongoDB and SQL Server.

EDUCATION

Future University in Egypt — B.Sc. in Computer Science, Cairo, Egypt *2022 – 2026*

University of Cincinnati — B.Sc. in Computer Science, USA *2022 – 2026*

CERTIFICATIONS & TRAINING

- **Data Science & AI Training** — Python, Pandas, NumPy, Scikit-learn, TensorFlow, Keras (Instant, 2026)
- **NVIDIA Deep Learning AI Fundamentals** — Deep Learning, CNNs, GPU Acceleration (2026)
- **Red Hat NLP & AI Solutions** — Natural Language Processing, Text Mining, AI Applications (2025)
- **Power BI for Data Analysis** — Dashboards, Data Visualization, Business Intelligence (2025)
- **Data Analysis with Python** — Data Cleaning, EDA, Visualization (2025)
- **Introduction to AI & Machine Learning** — Udemy (2024)
- **Python Programming (50 Hours)** — Impact Platform, Zewail University (2023)

LANGUAGES

Arabic: Native **English:** Professional Working Proficiency